



THÈSE DE DOCTORAT

DE SORBONNE UNIVERSITÉ

Modélisation des interactions aerosol-nuage dans les nuages en phase mixte en région arctique

Manuscrit soumis par

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Sciences de l'atmosphère

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Laboratoire Atmosphères,
Observations Spatiales (LATMOS)

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PrénomNOM Responsable des programmes composition atmosphérique et climat au CNES	<i>Invitée</i>
Jean-Christophe RAUT Professeur à Sorbonne Université	<i>Directeur de thèse</i>
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Abstract

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Keywords : Aerosol-cloud interactions – Biomass fires – – Pollutant emissions

Résumé

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Mots clés : Interactions aerosol-nuages – Feux de biomasse – Émissions de polluants

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Foreword

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¹Exemple de note de bas de page.

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Le comité de thèse.

Paul Z. and Julia K. and Annicka E.

Lea Raillard

Autres ami-es du LMD : Antoine B., Enora M., Léa B., Siméon

Niels Dutrievoz

Papa et Maman

Chloé et Etienne

Contributions to the scientific community

This side chapter is intended to list, and shortly describe, the various contributions of my thesis and side thesis activities to the scientific community. Here are presented the corpus of publications for which I contributed.

Peer-reviewed publications

Thesis research outputs

- Published in *Atmospheric Chemistry and Physics*: **Da Silva, A.**, Marelle, L., Raut, J.-C., Gramlich, Y., Siegel, K., Haslett, S. L., Mohr, C., and Thomas, J. L. (2025): How to trace the origins of short-lived atmospheric species: an Arctic example, *Atmospheric Chemistry and Physics*. DOI: <https://doi.org/10.5194/acp-25-5331-2025>
- Submitted in *A cool journal*:

Scientific collaborations

- Published in *Atmospheric Chemistry and Physics*: Lapere, R., Thomas, J. L., Favier, V., Angot, H., Asplund, J., Ekman, A. M. L., Marelle, L., Raut, J-C, **Da Silva, A.**, Wille, J. D., Zieger, P. (2024). Polar aerosol atmospheric rivers: Detection, characteristics, and potential applications. *Journal of Geophysical Research: Atmospheres*, 129, e2023JD039606. <https://doi.org/10.1029/2023JD039606>
- Published in *Tellus B: Chemical and Physical Meteorology*: Kojoj, J, Freitas, GP, Muilwijk, M, Granskog, MA, Naakka, T, Ekman, AML, Heutte, B, Schmale, J, **Da Silva, A.**, Lapere, R, Marelle, L, Thomas, JL, Melsheimer, C, Murray, BJ and Zieger, P. 2024. An Arctic Marine Source of Fluorescent Primary Biological Aerosol Particles During the Transition from Summer to Autumn at the North Pole. *Tellus B: Chemical and Physical Meteorology* 76(1): 47–70. DOI: <https://doi.org/10.16993/tellusb.1880>
- Submitted to ... : Antoine's paper
- Submitted to ... : Aymeric's paper
- Submitted to ... : Jessie's article about aerosols.
- Abstract at AGU ...: Megan Willis

-
- Abstract at AGU ...: Cort Zang

Conferences, colloquia, and workshops

2023

- EGU general assembly 2023, Vienna, oral presentation:
Da Silva, A., Marelle, L., and Raut, J.-C.: Seeking the origins of Arctic ice nucleating particles with FLEXPART-WRF, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-6392, <https://doi.org/10.5194/egusphere-egu23-6392>, 2023.
- SURFEIT kick-off meeting, *British Antarctic Survey* (BAS), Cambridge, UK, oral presentation.

2024

- Third QUESCENT workshop, EPFL, Lausanne, poster and oral presentations:
 - **Da Silva, A.**, Marelle, L., Lapere, R. and Raut, J.-C.: Are Primary Biological Aerosol Particles effective as reproducing Arctic INPs?, 3rd QUESCENT Workshop, Lausanne, Switzerland, 24–28 October 2024 (poster presentation)
 - *A short note about atmospheric species sources identification* (oral presentation)

2025

- INP Colloquium, ETH, Zurich, oral seminar presentation:
How to trace the emission sources of Ice Nucleating Particles: an Arctic example
- EGU general assembly 2025, Vienna, oral presentation:
Da Silva, A., Marelle, L., Lapere, R., and Raut, J.-C.: Modelling Ice Nucleating Particles from High-Latitude Sources to Reproduce Arctic In Situ Concentrations, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-16455, <https://doi.org/10.5194/egusphere-egu25-16455>, 2025.

International and national science projects

- Climate Relevant interactions and feedbacks: the key role of sea ice and Snow in the polar and global climate system (CRiceS).

The CRiceS project focuses on improving model predictions of the role of polar processes in the climate system that consists of the oceans, ice and snow cover, and the atmosphere. It is crucial to understand the role of the polar processes, such as feedback loops, in polar and global climate. One of the main ways scientists can improve our understanding of environmental change is to combine knowledge from different disciplines in a coordinated way.

My work contributed to the work package 2 (Improved ocean-ice/snow-atmosphere processes in models) and to the core theme 2 (Aerosols and Clouds).

- Cloud-aERosol inTeractions and their impActs IN The earth sYstem (CERTAINTY)

CERTAINTY aims to deliver the knowledge and models that provide improved confidence and representation of the role of cloud–aerosol-radiation interactions in climate and weather. This translates to better understanding and predictions of extreme events and facilitates planning climate mitigation and adaptation strategies for the good of the society.

My work contributed to the work package 3 (Improved parameterization of cloud-aerosol interactions).

- MPC2 ...

Teaching activities

Sorbonne Université

I took part in six different courses, teaching students from first year to master level, within two departments of the university (*Earth's sciences* and *Physics*). Below stands the detailed list of those courses.

Undergraduate

- *Terre, Climat, Environnement*: tutorial leading and elaboration of presentations topics for the new maquette of the course.
- *Le changement climatique et ses enjeux*: supervising of students talks about climate-society relations.
- Matlab modelling project: supervising of modelling projects for five groups of students in geology.
- Physique des milieux continus : charge des TDs et des résolutions de problème (RP).

Graduate

- *Statistics* : tutorial leading and project supervising in advanced statistics with R.
- *Rayonnement et télédétection* : supervising of the lidar practical work session.

Ecole nationale supérieure des études avancées (ENSTA) Paris

I have been tutorial leader for the transdisciplinary course named *Analyse systémique de l'Anthropocène* (Systemic analysis of the Anthropocene), for engineering school students.

European winter school ERCA 2025

I have been project group leader for six international PhD students. I supervised the group that focused on aerosol-cloud interactions. The objective I set up was to highlight the effect of ice nucleating particles on Arctic clouds thanks to principal component analysis of in situ measurements, and to empirical orthogonal functions computed on reanalysis data (ERA5 and CAMS).

Climactions IPSL

Working group research core activities

During the first two years of my PhD, I hosted the IPSL working group about the evolution of the observational activities of the climate sciences community. The group gathered researchers of various fields among the IPSL community. Our monthly meeting aimed to

Villarcieux Agreements and tool box

...

Transform'actions at LATMOS

...

Science communication and awarness raising

***Ma thèse en 180 secondes* 2023 contest**

I participated to the francophone contest *Ma thèse en 180 secondes*, where PhD candidates from every fields have 180 seconds to present their thesis to a non-specialist audience. I did the Sorbonne Université final. My appearance has been subtitled and can be found here: <https://www.youtube.com/watch?v=pGzU3e5XTz0>.

High school teaching

Interventions about global warming and climate sciences in high school classes (*2nde*, *1ère* and *terminales*). Atmospheric physics workshop with *terminales* students specialized in mathematics.

RADICAL

...

Lab life

...

I

Introduction to the Arctic aerosol and cloud physics

"Human beings, who are almost unique in having the ability to learn from the experience of others, are also remarkable for their apparent disinclination to do so."

– Douglas Adams

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I.1 What does the Earth get from the Sun

I.2 The Arctic atmosphere

I.2.1 Polar night and stability

I.2.2 Polar amplification

I.2.3 Modelling the Arctic climate

I.3 Clouds and microphysics

I.3.1 What is a cloud

The Earth's atmosphere contains at any moment not less than 10^{25} hydrometeors. The majority are liquid water droplets, and the rest is ice crystals (**cantrell_production_2005**). This number is to be put into perspective with the overall cloud coverage on the planet, which tends to be of 70% **<empty citation>**

I.3.2 Cloud formation and evolution

I.3.3 Cloud phase

Although the overall cloud ice content is small compared to the liquid content, ice crystals have a great influence on cloud evolution, precipitations and radiative budget.

I.3.4 Ice nucleation

I.4 Aerosols particles in the Arctic

I.4.1 Liquid and solid particles in suspension

I.4.2 Natural or anthropogenic?

I.4.3 Emission mechanisms

II

Methodological approach and techniques

"One does not simply compile WRF-Chem."

– Boromir of GitHub

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II.1 [Introduction]

II.2 Models and related tools

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III

Identification of ice nucleating particles emission sources

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III.2 How to trace the origins of INPs [first paper]

III.3 Natural sources of INPs in the Arctic

III.3.1 The MOCCHA 2018 campaign

III.3.2 The MOSAIC 2019-2020 campaign

III.3.3 The ARTOFMELT 2023 campaign

III.3.4 [What else?]

IV **Role of high latitude aerosols sources in the Arctic ice nucleating particles populations**

One does not simply compile WRF-Chem.

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IV.1 [Introduction]

IV.2 [Second paper]

IV.2.1 Methods

IV.2.1.1 WRF-Chem modelling setup

IV.2.1.2 High latitude dust emissions

The dust emissions rely on...

IV.2.1.3 Sea sprays emissions

IV.2.1.4 Observational data

IV.2.1.5 Ice nucleation modelling

IV.2.1.6 INP emission and activation

IV.2.2 Results

IV.2.2.1 First approach: in situ nucleation modelling

Coarse aerosols to INPs

IV.2.2.2 Emissions of modelled INPs

IV.2.3 Discussion

V **The effect of ice nucleating particles on phase partitioning and radiative properties of modelled Arctic clouds**

'''

– the author

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V.1 [Introduction]

V.2 [Third paper]

V.3

V.4

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RÉSUMÉ

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MOTS CLÉS

Interactions aerosol-nuages – Feux de biomasse – Émissions de polluants